

Text Analysis with R:

From raw data to AI assistant insights

Hyeseon Seo | UVA Library

StatLab

Room 152 & 153



Our Location Brown Library



Walk Through The Mural Area



Why Text Analysis?

1. Too much text to read: What are people talking about overall?

Government documents

Social media

Digital archives

Books, News, letters, speeches, transcripts...

2. Objectivity : Reduce subjective impressions

Text analysis provides evidence based on the entire corpus

3. Discover hidden patterns:

Recurring themes

Associations between words

Differences between groups

Sentiment trends

About this workshop

- A transcription : 2025 City of Charlottesville council meeting on ordinance on encamping and storage in property of the city.
- You will learn to:
 - Collect and clean transcript data
 - Five text analysis methods
 - Create basic visualizations
- We also look at how AI can support this work — and its limitations and ethics.

Three parts

- **Collect & Clean**

- Scrape the raw transcript from the web page into a tidy table.
- Tokenize, remove stopwords, subset to the ordinance, attach speaker names.

- **Analyze & Visualize**

- Explore the text and test pilot hypotheses across four groups.
- Faceted horizontal bar graphs.

- **AI support**

- Where and how—and its limitations and ethics

PART A

Collecting and cleaning the data

File 01 — Data collection

- **Can we scrape this?**
 - Check robots.txt, prefer an API, be a polite guest (identify yourself, throttle).
- **Look before you code**
 - Is the text in the page? Find the repeating pattern: transcript $\supset$ turns $\supset$ caption lines.
- **HTML $\rightarrow$ tidy table**
 - Use meaningful CSS selectors (li.speaker-chunk) to pull the 457 turns into rows.
- **Check & save**
 - Sanity-check the parse, save the raw HTML and a clean CSV.

File 01 — subset & clean

- Slice by time — keep only the three discussion periods on the ordinance.
- Rebuild turns — combine caption lines back into complete speaking turns.
- Attach identities — join a hand-coded speaker roster for names and stakeholder groups.
- Drop off-topic speakers (manually verified).
- **Output: ordinance.csv — a clean, labeled dataset ready to analyze.**

PART A (cont.)

Analyzing the transcript

File 02 — the question

- How do four stakeholder groups talk about the ordinance differently?
 - Police Chief — explains the proposed ordinance
 - City Council — deliberates and votes
 - Business representatives — public speakers for business
 - Public — other members of the public

Five methods

Each answers a different question about how they talked

- **Relative word frequency** — word rates per 1,000 words (fair across groups).
- **N-grams** — recurring 2–3word phrases (“affordable housing”).
- **Keyness** — words distinctive to one group vs. the rest.
- **KWIC** — keyword in context: read behind the numbers.
- **Sentiment** — dictionary-based tone by group and over time.

Basic visualizations

- Bar chart of top words by group (relative frequency).
- Keyness plot for one group's distinctive words.
- Sentiment by stakeholder, and across the three comment phases.
- **Goal: make the numbers readable.**

PART B

Using AI in text analysis

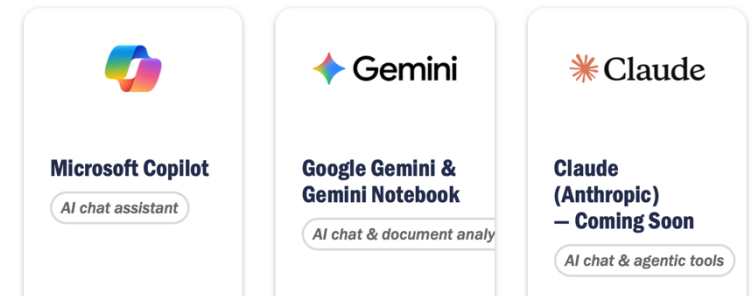
Five places AI fit in

Not to replace your thinking — to speed up the parts around it

- Brainstorm the topic and source
- Finding references
- Explore your data and scope the analysis
- Code support:
 - Write, fix and understand the code
 - Analysis, trouble shooting and visualization
- Writing

AI Tools

Explore new and popular AI tools available to all faculty, staff, and students through UVA ITS.



[AI tools that enable through UVA IT](#)

Brainstorm, scope, references

- **Brainstorm topic & source**

Relevant to the community, meaningful, free of content restrictions.

- **Explore the content**

AI can summarize the contents (e.g., transcripts),
identify themes, suggest connections among ideas, concepts, events.

- **Scope the analysis**

AI offered many options; we chose one theme — e.g., comparing four groups.

- **Search references**

AI also useful finding references, but verify each one (AI invents citations)

Reliable AI search tools grounded in peer-reviewed papers: [Consensus](#) and [LeapSpace](#) (UVA subscribes both!)

Coding support: how to ask

- This is AI as an assistant — not full agentic coding (Codex, etc.).
- **Give AI context — and the context is your data structure**
 - Screenshots are useful!
 - Lighter memory — a small screenshot or output of `str(data)` conveys the shape without a huge paste.
 - Safer — keeps the actual data out of the tool.
 - Ask precisely — the more specific, explicit prompt, the better the code
 -

Coding support: fixing & understanding

- **Troubleshooting**

- Screenshot your code AND the output together. Feed both.

- **Understand the code**

- When you are unsure about specific code, paste the whole script (context matters!) and ask what each line does.
- Sometimes ask for an example what a specific part does.

- **Visualization**

- Which chart best fits your data

Writing

- **Polish the writing**

- Tighten wording and clarify; keep your own argument and voice.

- **Manage citations**

- Format citations in different styles (e.g., APA, Chicago,) and organized reference in the appropriate order.

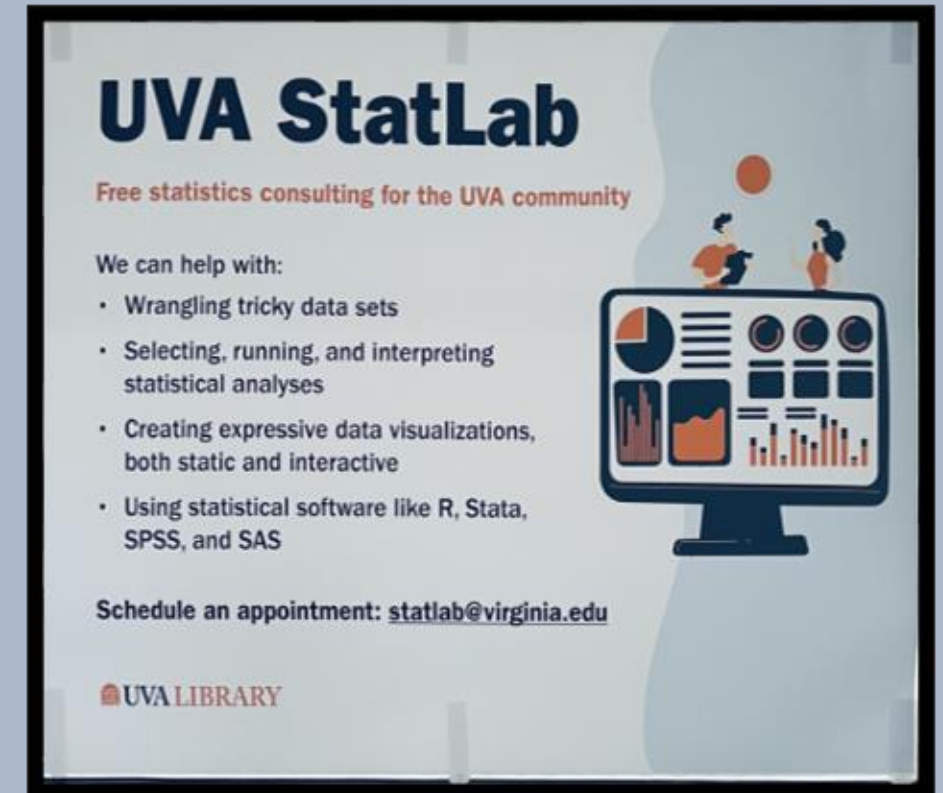
Limitations & ethics

- AI invents citations — verify everything.
- Understand what your codes does.
- Protect your data — share structure, not raw data.
- Check the AI rules that apply to you — your institution's and your field's (UVA [AI Guidelines](#))
- **AI assists; you decide and stay accountable.**

Thank you — questions?

StatLab

- Walk-in hours:
 - Tue 1–3
 - Wed 2–4
 - Thu 10–Noon
- statlab@virginia.edu
- <https://library.virginia.edu/data/statlab>



Appendix

AI Prompt Examples

1. Brainstorm topic & source

"I want to do a text-analysis workshop using something locally relevant and free of content restrictions. Suggest 3–4 public, scrapeable text sources relating to Charlottesville community. For each, note whether it's likely server-rendered HTML and whether a robots.txt would allow scraping."

2. Explore the content

"Here's the transcript of the Sept 2, 2025 Charlottesville City Council meeting Summarize the main topics, then identify the distinct discussion periods and who the key stakeholder groups are. Suggest 2–3 themes worth analyzing quantitatively."

3. Scope the analysis

"This meeting covers many topics over 6 hours. I want to narrow to one theme, give me possible themes."

4. Search references (verify each one!)

"I'm comparing how stakeholder groups use language in a public-meeting transcript. Recommend peer-reviewed methods papers on relative word frequency, keyness (chi-squared), and sentiment lexicons for small corpora."

5. Give AI context = your data structure (use screenshots or str())

"Here's the shape of my data [paste output of str(data) or a screenshot of head(data)]. Keep this structure in mind for the code I ask for next."

Then ask for the code : "The speaker column is a mix of real names ('Michael Kochis') and automatic labels ('SPEAKER_45'). Some rows have NA. Write a code: fill NA with the speaker label, replace any name with 'unidentified' (case-insensitive) with the speaker label, and otherwise keep the name."

6. Ask specifically

Vague: "Analyze this transcript."

Precise: "Compare the four stakeholder groups in this City Council transcript. Identify differences in relative word frequency, key terms, and sentiment, and summarize the main patterns."

7. Troubleshooting (feed code + output together)

"[Screenshot of my unnest_tokens code AND the error/output below it.] fix it."

8. Understand the code "Here's my full tokenizing script [paste the code block]. Walk me through what each line does, especially anti_join(stop_words), the filter(str_detect(word, "^[a-z]+\$")), and why case_when() is used to collapse 'visitor'/'visitors'."

Or focus part on one part you don't understand more specifically

"Give me an example of what reorder_within() does to a two-group dataset — show the before and after."

9. Visualization (which chart fits)

"I have three per-group measures — number of speakers, turns per person, and median words per turn — for four stakeholder groups. What chart type best shows all three at once for comparison across groups?"

10. Polish the writing (keep your voice)

"Here's my interpretation paragraph for the relative-frequency chart [paste your draft]. Tighten the wording and fix flow, but keep my argument and voice" and fact check.

11. Manage citations

"Citate these three references to APA 7th edition, then give me the same list in Chicago author-date. Sort each list alphabetically by author surname."